ult2s4.py: Porting Ultimate Animations to Smash 4

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Overview

ult2s4.py turns Smash Ultimate animations into Smash 4 (Wii U) animations and writes them straight into the fighter's motion .pac files. It is the reverse of JoshuaSawagi's S4-Anims-to-Ult, as one Python script plus an optional point-and-click GUI.

- **Input:** Ultimate animations exported from StudioSB as Maya `.anim` files, plus the Smash 4 fighter's `model.vbn` and its `motion/body/*.pac` files.
- **Output:** copies of every .pac with the chosen motions replaced, the loose `.omo` files, and a `report.txt` listing what changed.
- **Untouched:** every other animation in the .pac files stays byte-for-byte identical to the input.

Tested on Daisy's Ultimate animations onto Smash 4 Peach. Daisy's dash comes out within 0–5° of Peach's stock dash on every body bone.

Get started

Everything ships in `Ult2S4.zip`. Unzip it anywhere, keep the files together, and run the one-time setup.

File	What it is
`Ult2S4 GUI.bat`	Double-click to open the GUI
`ult2s4_gui.py`	The GUI; must sit next to `ult2s4.py`
`ult2s4.py`	The converter; also runs from the command line
`install_numpy.bat`	One-time setup that installs numpy
`README.txt`	Short version of this guide

1. Install Python 3 from python.org and tick **Add python.exe to PATH** in the installer.

2. Double-click `install_numpy.bat`. It ends with a line saying numpy is installed.
3. Double-click `Ult2S4 GUI.bat`.

Requirements

You need Python 3 with numpy, StudioSB, and the files below from both games.

What	Where it comes from	Notes
Python 3	python.org	Any recent version; tick **Add python.exe to PATH**
numpy	`install_numpy.bat` from the zip, or `py -m pip install numpy`	The only extra library
Tkinter	Included with the python.org installer	Only needed for the GUI
StudioSB	Smash Ultimate modding tool	Opens Ultimate models and exports animations as `.anim`
Ultimate fighter files	Extracted from Ultimate (`fighter/<name>/model/body/c00` and `fighter/<name>/motion/body/c00`)	The model so StudioSB has a skeleton, and the `.nuanmb` animations
Smash 4 `model.vbn`	`fighter/<name>/model/body/c00/model.vbn` (any slot with the stock skeleton)	Gives the bone IDs and rest positions
Smash 4 motion .pac files	`fighter/<name>/motion/body/` from your Sm4shExplorer extract or workspace	main.pac, lhand.pac, rhand.pac and the swing pacs (e.g. peachsw1.pac)

Step 1: Export the Ultimate animations

Each Ultimate animation you want becomes one `.anim` file, exported from StudioSB with every bone included.

1. Open the Ultimate fighter's model in StudioSB (the `model/body/c00` folder), so the skeleton is loaded.

2. Load the `.nuanmb` animation from `motion/body/c00` (for example `a00wait1.nuanmb`).
3. Export it as a Maya `.anim` file. Include all bones and keep every frame.
4. Name the file after the motion, like `a00wait1.anim` or `a02run.anim`. The script reads the Smash 4 motion name from the file name.
5. Repeat for each animation, and put all the `.anim` files in one folder.

Rotations in the `.anim` can be in degrees or radians; the script reads the file's own unit setting and converts.

Step 2: Gather the Smash 4 files

The script needs the fighter's skeleton file and one folder holding its motion .pac files.

- **Skeleton:** copy `model.vbn` from `fighter/<name>/model/body/c00`. A custom costume's VBN works too, as long as it uses the stock skeleton.
- **Motion pacs:** copy everything in `fighter/<name>/motion/body/` into its own folder, such as `pacs/`. For Peach that is `main.pac`, `lhand.pac`, `rhand.pac`, `peachsw1.pac` and `peachsw2.pac`.
- **Starting point:** use the stock pacs from your extract for a clean start. Use your workspace pacs instead if they already hold animations you want to keep; only the motions you name get replaced.
- **Folder contents:** keep only that fighter's .pac files in the folder. Every .pac in it is read and written back out.

Sm4shExplorer's workspace folders are nested deep, so copying these files to a short path (like a folder in Downloads) makes the command easier to type.

Step 3: Convert with the GUI

The GUI does the whole job from one window: pick files, convert, read the report, install with a backup.

1. **Smash 4 files:** use Browse to pick the skeleton (`model.vbn`), the motion folder holding the .pac files, an output folder, and optionally a Stock reference folder. The output folder must differ from the motion folder. The stock reference is a copy of the fighter's untouched original .pac files; set it once your motion folder already holds converted moves, so the dress is always measured against the real Smash 4 one.

2. **Add animations:** click **Add .anim files...**. Each file gets a slot from its name, matched to the real motion names in `main.pac`; the count of slots found shows bottom right.
3. **Change a slot:** double-click a row and pick from the list of the fighter's motions.
4. **One animation, two slots:** select a row, click **Add same file to another slot**, then double-click the copy to set its slot.
5. **Options:** leave Skirt on **auto** and keep **Push skirt out where legs poke through** ticked unless you want to compare.
6. Click **Convert**. The log shows the same report as `report.txt`, and the status line says when it is done.
7. Click **Install into motion folder...**. Your current .pac files are copied to `backups\<date_time>\` next to the script first, outside the workspace, so they never end up in the patch.
8. Rebuild the patch in Sm4shExplorer and test.

Theme. Options → Theme switches between Windows default, Light, Dark, Midnight, Forest, Peach and Daisy. **Accent color...** sets any highlight colour, and **Reset** goes back to the theme's own. The window remembers your folders, list, theme and accent between uses in `ult2s4_gui_settings.json`.

If a slot you typed isn't in `main.pac`, the GUI warns you before converting; those rows are skipped.

Or: run it from the command line

One command converts any number of animations at once. Open a Command Prompt in the folder that holds `ult2s4.py` and run:

```
python ult2s4.py --vbn "C:\port\model.vbn" --pacs "C:\port\pacs" --out
"C:\port\out" "C:\port\anims\a00wait1.anim" "C:\port\anims\a02run.anim"
```

Ready-made for Peach in your setup: convert into an `out` folder next to the script, then copy the results in. Paste these into Command Prompt one at a time and change the `.anim` name at the end:

```
cd /d "C:\Users\tommy\Downloads\decompiled 1.8.2\Claude Stage
Projects\Ult2S4"
py ult2s4.py --vbn "..\peach_check\c09\model.vbn" --pacs
"C:\Users\tommy\OneDrive\Desktop\Wii U Modding\sm4shexplorer
0.07.4\workspace\content\patch\data\fighter\peach\motion\body" --out out
"..\a02run.anim"
copy /y out\*.pac "C:\Users\tommy\OneDrive\Desktop\Wii U
Modding\sm4shexplorer
0.07.4\workspace\content\patch\data\fighter\peach\motion\body\"
```

If `py` isn't recognised, use `python` instead. Unlike the GUI's Install button, `copy` makes no backup.

Argument	Required	What it does
`--vbn`	Yes	The Smash 4 `model.vbn` from Step 2
`--pacs`	Yes	The folder holding the fighter's motion .pac files
`--stock`	Recommended	A folder with the fighter's untouched original .pac files, used as the skirt and leg reference. Defaults to --pacs; set it once --pacs already holds converted moves
`--out`	Yes	Where the new .pac files, .omo files and `report.txt` go (created if missing)
`--prefix`	No	The motion name prefix, e.g. `FitPeachBody`; read from `main.pac` automatically
`--skirt`	No	`auto` (default), `stock`, `anchored`, `neutral` or `ult`; see Skirt handling below
`--no-pushout`	No	Turns off the skirt push-out
`file.anim`	At least one	An animation to convert; its motion name comes from the file name
`file.anim=MotionName`	Optional form	Puts that animation into a different motion slot

Skirt handling (`--skirt`)

The Peach/Daisy skirt is built differently in Ultimate, so the script picks a skirt source per animation. The default, `auto`, chooses for you and logs its choice at the top of each motion in `report.txt`.

Mode	What the skirt does	When to use it
`auto` (default)	`stock` when the converted legs match the stock move (same length, legs within 8°); stock + push-out when the length matches but the legs differ; otherwise anchored + push-out, or `neutral` + push-out when Ultimate's baked skirt keys jump more than 60° in one frame	Almost always
`stock`	Keeps Smash 4's own skirt motion for that move	Moves Ultimate reused from Smash 4, like dash and run
`anchored`	Ultimate's sway around Smash 4's natural skirt shape	New moves, like Daisy's idles
`neutral`	Holds Peach's average dress pose still; only the push-out moves it	Attacks whose Ultimate skirt keys snap around, like down tilt (auto picks it)
`ult`	Ultimate's skirt rotations as they are	Only for comparison; this is what caused leg clipping and a raised back hem

Push-out. With `anchored` skirts, and stock skirts under legs that move differently, the script also checks every frame for a knee, shin or shoe poking through the dress. Where one does, it swings the nearest skirt panels outward just enough to cover the leg, easing in and out over a few frames. It never lifts a hem above the knees in idles, and in moves that use Smash 4's skirt it lifts the hem at most half a unit, leaving the hem overlap Peach's own move already has. For moves whose legs differ from Peach's version by more than 45°, the dress starts from her standing dress (A00Wait1), and the hem never rises above where that dress reaches. Big kicks will show some leg rather than flip the skirt. Ultimate simulates the skirt in game, so the skirt keys baked into an attack's export can jump over 100° in one frame; when they do, auto switches to neutral. `report.txt` lists the frame counts before and after for each motion. Add `--no-pushout` to turn it off.

Motion names

The file name sets the slot: `a00wait1.anim` becomes `A00Wait1`, `a02run.anim` becomes `A02Run`. Matching ignores upper and lower case.

When the Ultimate and Smash 4 names differ, or you want an animation in a second slot, add `=` and the Smash 4 name. The same file can be listed more than once:

```
... a00wait2.anim a00wait2.anim=A00Wait4
```

That puts Ultimate's Wait2 into both Smash 4's Wait2 and Wait4.

What it prints

One line per motion per .pac, also saved to `report.txt`:

```
A02Run: main.pac FitPeachBodyMainA02Run.omo  33 frames (stock 33), 34 bones,
8 kept from stock
```

- **frames (stock N):** the new length next to the original Smash 4 length.
- **kept from stock:** bones Smash 4 animates that Ultimate has no match for; they keep their original motion, stretched to the new length.
- **has no ... skipped:** that .pac has no motion by that name. Check the spelling or use `=MotionName`.

Step 4: Install and test

Copy the new .pac files into the workspace, rebuild the patch, and check the motions in game.

In the GUI, **Install into motion folder...** does steps 1 and 2 for you, with the backup in `backups\<date_time>\` next to the script. From the command line, do them by hand:

1. Back up the fighter's current files in `workspace/content/patch/data/fighter/<name>/motion/body/`.
2. Copy every `.pac` from your `--out` folder into that folder, replacing the old ones.
3. Rebuild the patch in Sm4shExplorer as you normally do.
4. Start a match in Cemu and watch each converted motion: idle for the waits, hold a direction for dash and run.

The motion .pac files are shared by every costume of the fighter, so the new animations show on all slots.

To undo: copy the original .pac files back from your backup, or from `extract/data/fighter/<name>/motion/body/` for stock, and rebuild.

How the conversion works

For each animation the script runs the same six steps, then writes one .omo per .pac.

1. **Read the keys.** It takes the `.anim` keyframes 1 to N. StudioSB sets the end time to N+1, so N matches the frame count Smash 4 uses for the same move.
2. **Rename bones.** Ultimate names become Smash 4 names using the reverse of the original tool's 288-pair table, plus `HaveL/R`, `Throw`, `Trans`, `Rot`, `Hip` and the Peach/Daisy swing bones.
3. **Fold the extra bones.** Ultimate adds `ClavicleC` above the shoulders and `LegC` above the legs. Their motion is merged back into `LShoulderN` / `RShoulderN` and `LLegJ` / `RLegJ`, so the limbs end up in the same place.
4. **Convert rotations.** Euler angles become quaternions using the order R = Rz · Ry · Rx, the order both games share.
5. **Handle rig differences.** Bones whose Ultimate rig was rebuilt (the Peach/Daisy skirt) keep Smash 4's bone lengths. Their motion comes from the skirt mode (Smash 4's own skirt, or Ultimate's sway on Smash 4's skirt shape), then the push-out swings panels clear of the knees and shoes. Face bones are skipped, since Smash 4 body animations don't animate them.
6. **Split by .pac.** Each .pac gets exactly the bones its stock .omo animated: body bones to main.pac, fingers to lhand/rhand.pac, swing bones to the swing pacs. A bone Ultimate lacks keeps its stock motion, stretched to the new length.

The .omo files use the same 16-bit keyframe layout as Smash Forge, which already ran in game for the stage work. The .pac writer rebuilds stock .pac files byte-for-byte, so nothing else in them shifts.

Limits and troubleshooting

Idles and movement port cleanly. Attacks, skirts and other rigs need care.

Problem	Cause	What to do
Report says `has no ... skipped`	That .pac has no motion with that name	Check the spelling, or name the slot with `file.anim=MotionName`
Nothing changed in game	The patch wasn't rebuilt, or the pacs went to the wrong folder	Copy to `motion/body/` and rebuild in Sm4shExplorer
Hitboxes don't line up with an attack	Timing now follows Ultimate, but Smash 4's `game.bin` still uses the old frames	Stick to idles and movement, or retime that move's script
Animation is longer or shorter than before	Length follows Ultimate	Fine for loops and idles; the game plays the whole thing
Skirt or hair moves differently	Ultimate rebuilt those swing bones	Keep --skirt auto and the push-out on; report.txt shows poke-through frames before and after
Arms or legs twisted badly	The two skeletons don't match for this fighter	Check bone names; see the next section
Many bones `kept from stock`	Ultimate bone names didn't map	Add the missing names to the table (next section)
Script fails to read a `.anim`	Export missing keyframes or bones	Re-export with all bones and every frame

- **Every costume changes.** Motion pacs are shared by all slots of the fighter.
- **Face expressions don't carry over.** Smash 4 handles faces with separate expression files, not body animations.

Using it for other fighters

Body and finger bones already map for every fighter the original S4-Anims-to-Ult covers. Swing bones (hair, capes, tails, skirts) are named per fighter, so each new fighter needs its own lines added.

Find the names.

- **Ultimate:** open a `.anim` in a text editor. Each bone appears in lines like `anim rotate.rotateX rotateX S_Hair1 ...`.
- **Smash 4:** open the fighter's `model.vbn` in Smash Forge and read the bone list. Swing bones look like `SWG_Bhair0__swing`.

Add the lines. Near the top of `ult2s4.py`, under the existing hair line, add one line per bone:

```
U2S['S_Cape1'] = 'SWG_Cape0__swing'
U2S['S_Cape2'] = 'SWG_Cape1__swing'
```

Those cape names are made-up examples; use the real ones you found.

- **Rig rebuilt in Ultimate?** If the bone chain has different lengths in Ultimate, also add it to `ROT_ONLY` so Smash 4's lengths are kept: `ROT_ONLY.add('S_Cape1')`.
- **Extra in-between bone?** If Ultimate inserts a bone that Smash 4 doesn't have, add it to `FOLD` as `child: inserted parent`, the way `ClavicleL: ClavicleC` is set up.

Run the script and read `report.txt`. A drop in `kept from stock` means the new names mapped.

Worked example: Daisy onto Peach

Seven Peach motions were replaced with five of Daisy's Ultimate animations in one run. Daisy has only three idles, so her Wait2 and Wait3 also fill Peach's Wait4 and Wait5.

```
python ult2s4.py --vbn peach_check\c09\model.vbn --pacs pacs --out out_peach
a00wait1.anim a00wait2.anim a00wait3.anim a00wait2.anim=A00Wait4
a00wait3.anim=A00Wait5 a02dash.anim a02run.anim
```

Peach motion	Daisy file	New frames	Stock frames
A00Wait1	a00wait1.anim	236	101
A00Wait2	a00wait2.anim	94	240
A00Wait3	a00wait3.anim	130	270
A00Wait4	a00wait2.anim	94	190
A00Wait5	a00wait3.anim	130	125

Peach motion	Daisy file	New frames	Stock frames
A02Dash	a02dash.anim	35	35
A02Run	a02run.anim	33	33

- **Accuracy:** Daisy's dash matched Peach's stock dash within 0–5° on the body bones, since Ultimate reused that animation.
- **Loops:** the idles start and end on the same pose, so they loop without a pop.
- **Files:** the script, these pacs and the report are saved in `Claude Stage Projects\Ult2S4\`.

Skirt results

The push-out cleared almost every frame where a knee or shoe came through the dress.

Motion	Skirt mode	Poke-through frames before	After	Highest hem
A00Wait1	anchored + push-out	226 of 236	0	below the knees
A00Wait2	anchored + push-out	94 of 94	20	below the knees
A00Wait3	anchored + push-out	98 of 130	13	below the knees
A00Wait4	anchored + push-out	94 of 94	2	below the knees
A00Wait5	anchored + push-out	128 of 130	3	below the knees
A02Run	stock (legs within 3.7°)	—	—	Peach's own
A02Dash (`a02dash_v2.anim`)	stock + gentle push-out (legs 15.8° off)	26 of 35 beyond stock	17	about Peach's own

Motion	Skirt mode	Poke-through frames before	After	Highest hem
C02AttackLw3 (down tilt)	neutral + push-out (legs 66° from Peach's move; Ultimate skirt jumps 161°)	34 of 34	32 (worst 5.2, Peach's own 7.3)	below the knees (was above the hips)

The dash was later replaced with `a02dash_v2.anim`. The first push-out flipped its skirt up (a panel swung up to 148°). The push-out now limits how far it lifts the hem, so this dash keeps Peach's own flare; the remaining frames are feet at the hem.

Currently installed: the original Daisy dash (`a02dash.anim`) is back in place, with Peach's own skirt (legs within 1.4° of her stock dash, no push-out).

Version history

Date	Version	Change
2026-10-31	Length mismatch fix	Fixes the "operands could not be broadcast together" crash when a motion's reference files differ in length (victory face vs body, hair in down tilt and throws). Victory poses with dress files take the standing dress from motion\body when the reference has none
2026-10-31	Victory poses	Converting into motion\bodyresult works: face.pac is left as is (it no longer triggers "--skirt stock needs the same frame count"), and the dress step is skipped there because victory poses have no skirt file
2026-10-31	Down tilt fix	Stock reference folder (--stock) in GUI and command line; neutral skirt mode picked automatically when Ultimate's skirt keys jump; hem height limit for moves unlike Peach's. Idle, dash and run output unchanged

Date	Version	Change
2026-10-30	Push-out fix	Hem-lift limits stop the skirt flipping up (dash); idles keep the hem below the knees
2026-10-30	GUI themes	Theme dropdown (7 themes), custom accent colour, dark Windows title bar
2026-10-30	GUI	Point-and-click window, slot list from `main.pac`, install with automatic backup, remembered settings; zip with launcher and numpy installer
2026-10-30	Auto skirt rule	Same length but different legs now uses Smash 4's skirt timing + push-out
2026-10-30	Push-out	Skirt panels swing clear of knees and shoes; `--no-pushout` to disable
2026-10-30	Skirt modes	`--skirt auto/stock/anchored/ult`; fixes the raised back hem
2026-10-30	First release	Ultimate .anim → Smash 4 .omo inside the fighter's .pac files